

BTCSCS209**IV Semester Examination, May – June 2022****B.Tech. (CSBS-TCS)****Operating Systems**

Choice Based Credit System (CBCS)

Time: 3 Hrs.**Maximum Marks: 60****Minimum Pass Marks: 24**

- Note:**
1. All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.
 2. From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
 3. Draw neat diagram, wherever necessary.
 4. Assume suitable data wherever necessary.

- Q.1(A)** Explain various types of Operating System. **03**
- (B)** Discuss different services provided by Operating System in details **03**
- (C)** What are the different fields of PCB? Explain Context Switching with Triggers. **06**

OR

Explain life cycle of thread. Write the Difference between Thread and Process.

- Q.2(A)** Define Process Scheduling. List any five scheduling objectives. **03**
- (B)** Compare Long term Scheduler, Short term scheduler and Mid-term scheduler. **03**
- (C)** Consider the set of 5 Jobs whose arrival time and burst time are given below:

Jobs	Arrival Time	Burst Time
J ₁	3	1
J ₂	1	4
J ₃	4	2
J ₄	0	6
J ₅	2	3

Consider the CPU scheduling algorithm Shortest Job First (SJF) with Pre-emptive and Non pre-emptive mode. Find the sum of Average waiting time of SJF Pre-emptive and Non pre-emptive respectively. **06**

OR

Explain following term:

1. EDF
2. MLFQ Scheduling
3. Round Robin with example

- Q.3(A)** What is Producer Consumer Problem? Write solution using Semaphore. **03**
- (B)** What is Deadlock? Brief the necessary and sufficient conditions for Deadlock. **03**

(C) Discuss the following term in brief:

1. Test and Set Lock hardware solution
2. Violation of Circular Wait
3. Race Condition

06

OR

Consider following snapshot:

Process	Max	Allocation	Available
	A B C D	A B C D	A B C D
P0			
P1	6 0 1 2	4 0 0 1	3 2 1 1
P2	2 7 5 0	1 1 0 0	
P3	2 3 5 6	1 2 5 4	
P4	1 6 5 3	0 6 3 3	
P5	1 6 5 6	0 2 1 1	

Using Banker's algorithm, answer the following questions:

1. How many resources of type A, B, C, D are there?
2. Find Need Matrix
3. Is the system being in safe state? If it is, find the safe sequence.

Q.4(A) Explain Internal and External fragmentation problem with suitable example using Fixed size partitioning scheme.

03

(B) Write the difference between paging and segmentation.

03

(C) Discuss Belady's Anomaly Problem? Consider the following page reference string 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 for a memory with 3 frames. How many page fault would occurs if FIFO page replacement algorithm is used?

06

OR

Consider a Program size is 4GB, physical address space is 64 MB and size of the page is 4KB. Find the following:

I) No. of Pages, II) No. of Frames, III) No. of Entries in Page table, IV) Size of the Page table

Q.5(A) Explain Free space management techniques.

03

(B) If HDD size is 32 MB and Block size is 1 KB. Find the size of Bit Vector and how many blocks are required to store Bit Vector.

03

(C) What is directory structure of the file system? Explain directory implementation methods.

06

OR

A disk contains 200 tracks (0-199). Request queue contains tracks no. 82, 170, 43, 140, 24, 16, 190 repeatedly. Current position of Read-Write head is 50. Calculate total no. of tracks movement by R/W head using SCAN algorithm. Consider direction of R/W head is toward larger value.
